INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME			
CLASS		INDEX NUMBER	

CHEMISTRY **9647/02**

Paper 2 Structured Questions **19 Sept 2014**

2 hours

Candidates answer on the Question Paper

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.
A Data Booklet is provided.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	**12**
2	**13**
3	**15**
4	**18**
5	**14**
Significant figures	
Handwriting	
Total	**72**

This document consists of **18** printed pages.

For Examiner's Use

Answer ALL questions on the spaces provided.

1 Planning (P)

Early in the nineteenth century, Michael Faraday observed that the masses of the substances produced in an electrochemical reaction depend on the amount of current transferred at the electrode. The electrical charge carried by one mole of electrons is One Faraday in his honour.

Zinc is one of the most extensively used metals in the surface finishing industry. Zinc plating provides a soft and protective anti−corrosive barrier which acts as a sacrificial coating that prolongs the life of the underlying metal of items such as screws, bolts, fasteners and automotive parts.

You are to plan an experiment to determine the value of Faraday constant from the measurements made during electroplating zinc on the surface of a screw using a suitable electrode.

(a) Write the half equations with state symbols for the reaction at the anode and cathode.

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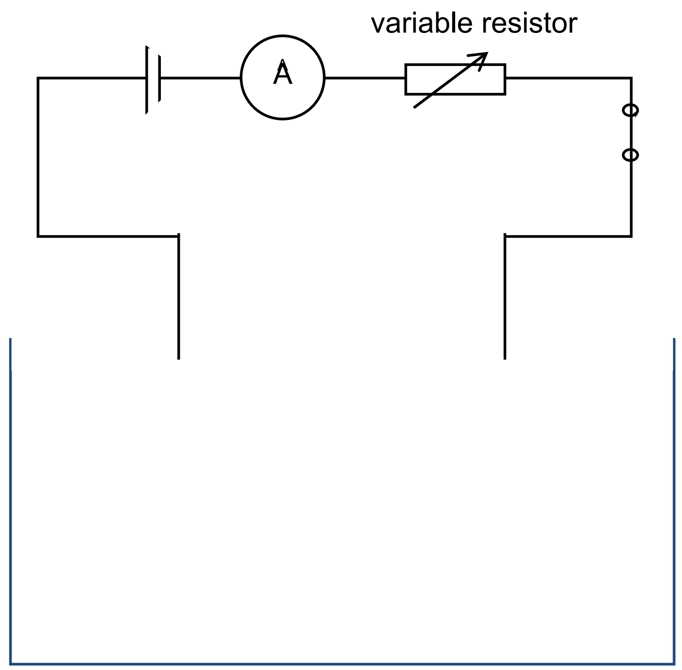
[1]

(b) Complete and label the diagram below to show the experimental set up for coating a screw with zinc metal by electrolysis. Suggest a suitable concentration for the zinc sulfate electrolyte.



[1]

(c) Write a detailed plan to prepare a stock solution of 500 cm^3 of 1 mol dm^{-3} of Zn^{2+}(aq) solution from a bottle of zinc sulfate solid (molar mass 161.5 g/mol) provided.

Your plan should include:

- all calculations to obtain the quantity of chemicals added
- apparatus or measuring device used
- all precautions taken to ensure accuracy

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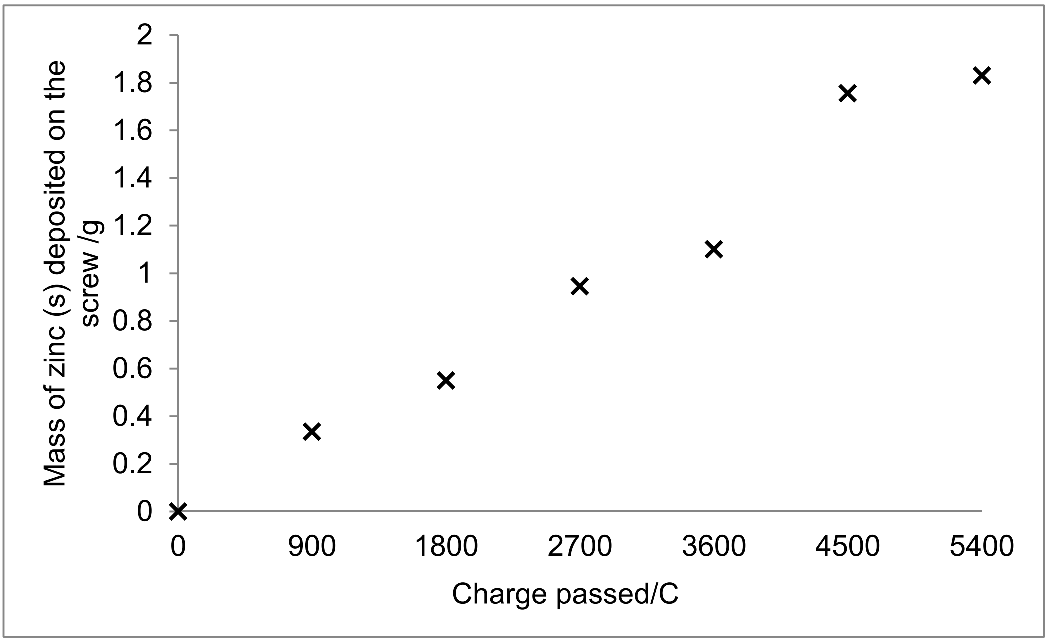
[4]

(d) Using the stock solution as electrolyte, the student conducted zinc electroplating of the screw. Before the screw is placed into the electrolyte, it was cleaned with the emery paper and weighed.

The power source was connected to the ammeter, rheostat and electrolytic cell as shown in **(b)** using crocodile clips. With the rheostat set to maximum resistance, the d.c. supply was switched on and the rheostat was adjusted until the current is 0.3 A. This current was kept constant at 0.3 A for 50 min. At the end of the 50 min, the zinc electrode was removed and rinsed with distilled water followed by propanone. It was then allowed to dry before it was weighed again.

The experiment was repeated for another 6 times at 50 min intervals. After mathematical manipulation, the mass of Zn(s) deposited on the screw was calculated at every specific time interval.

The student plotted the data on the graph of mass of Zn(s) deposited against charge passed through the circuit. Based on the plotted points, draw a best−fit line on the graph below.



[1]

(e) With reference to **(d)**, circle the anomalous point. Suggest an explanation for the anomaly.

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[2]

(f) The student calculated the value of the gradient according to the graph in **(d)** as 3.38×10^{-4} g C^{-1}. Calculate a numerical value of the Faraday constant.

[2]

(g) Suggest a reason why the percentage yield for electroplating will not reach 100%.

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[1]

[Total:12]

For Examiner's Use

2 Phenyl benzoate can be prepared by shaking phenol with benzoyl chloride in alkaline solution. The products appear as a solid and the pure compound can be obtained by dissolving in hot ethanol and cooling.

Preparation of impure phenyl benzoate

1. Place 2.0 g of powdered phenol and 90 cm^3 of concentrated sodium hydroxide in a round bottom flask.
2. Transfer 40 cm^3 of 58 g dm^{-3} benzoyl chloride into the round bottom flask and connect it to a reflux condenser.
3. Gently boil the mixture for about 30 minutes, shaking the flask gently from time to time.
4. Cool the flask under cold, running water.
5. Filter the crude product using suction filtration. Use a spatula to break up the lumps of ester on the filter paper, being careful not to puncture the filter paper.
6. Pour cold water over the crude product.

(a) Suggest why concentrated sodium hydroxide is added to phenol in step **1** before benzoyl chloride is added.

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[2]

(b) Write a balanced equation for the reaction between phenol and benzoyl chloride.

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[1]

(c) By using the information given above, one of the reagents, phenol or benzoyl chloride will be present in an excess in this preparation.
Use the data above to determine, by calculation, which reagent is in an excess.

[3]

For Examiner's Use

(d) Draw a fully labelled diagram of the set up for the preparation of the impure phenyl benzoate.

[2]

(e) The reaction mixture is heated for 30 minutes in step **3**.
Why is this process necessary for the preparation of many covalent organic compounds?

Explain your answer.

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[2]

(f) Cold water is added to the crude product in step **6.** Explain the function of the cold water.

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[1]

(g) Provide **two** possible reasons why benzoyl chloride is used to prepare phenyl benzoate instead of benzoic acid.

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[2]

[Total: 13]

For Examiner's Use

3 **(a)** A sample of copper contains the two isotopes ^{63}Cu and ^{65}Cu only. An experiment is conducted to find the relative atomic mass of this sample and is found to be 63.9.

Suggest why the relative atomic mass stated above differs from the value obtained from the Periodic Table.

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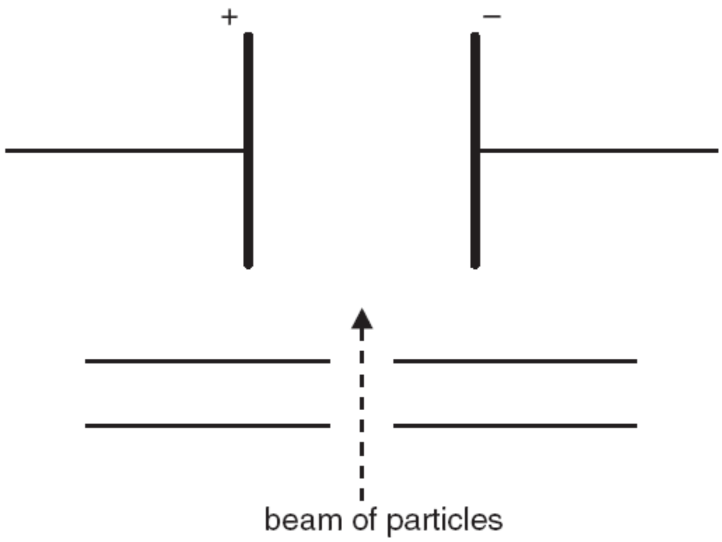
.... [1]

(b) When separate beams of $^{63}Cu^{2+}$ and $^{127}I^{-}$ are passed through an electric field in the apparatus below, they behave differently.



(i) Sketch on the diagram above how the paths taken by beams of $^{63}Cu^{2+}$ and $^{127}I^{-}$ in the presence of electric field.

(ii) Given that the angle of deflection of the $^{63}Cu^{2+}$ beam is +7.0°, calculate the angle of deflection of the $^{127}I^{-}$ beam, correct to 1 decimal place.

[4]

(c) The elements **B** and **C** can be one of the following: Na, Mg, A*l*, Si and P.

Element **B** has a chloride and an oxide which react vigorously with water to form solutions containing strong acids.

Element **C** has a crystalline solid oxide with a very high melting point. This oxide is classified as an acidic oxide but it is not soluble in water.

Identify the elements **B** and **C**. Explain the observations with the help of relevant balanced equations with state symbols where necessary.

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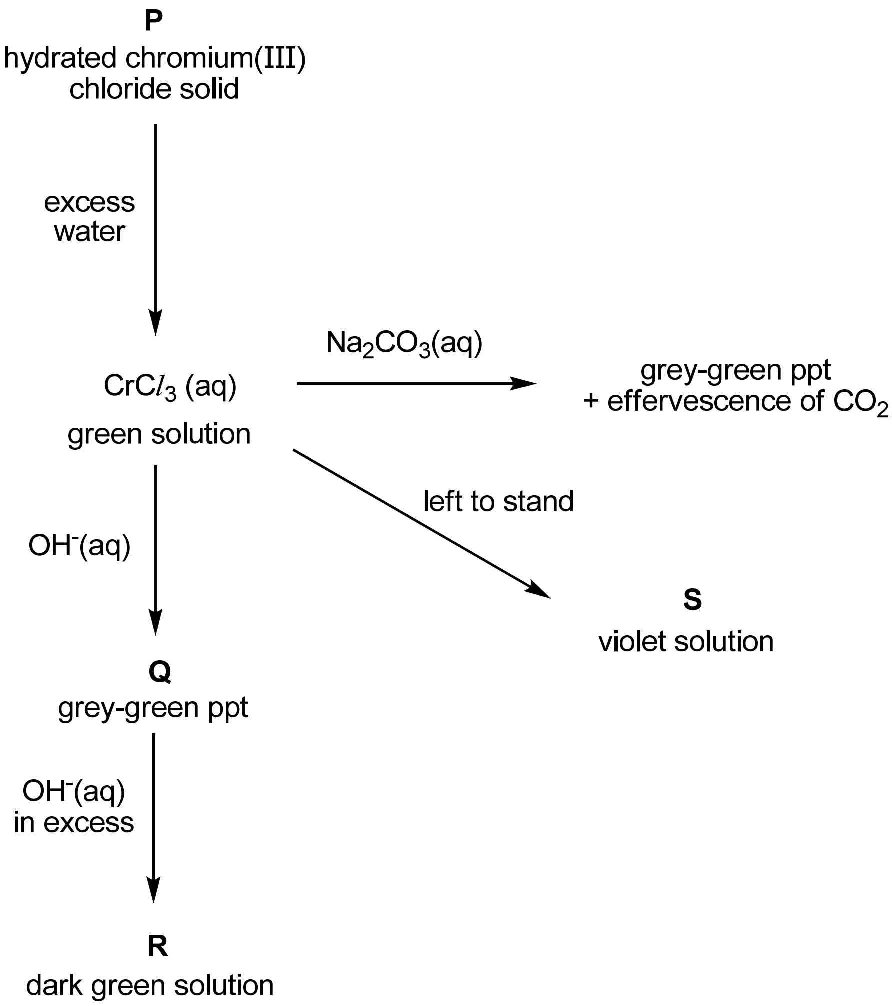
[4]

(d) The following are some reactions of chromium(III) chloride.



(i) Explain why carbon dioxide is evolved when Na_2CO_3(aq) is added to aqueous chromium(III) chloride. Include any relevant equations with state symbols in your answer.

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(ii) Identify the grey−green precipitate **Q**.

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(iii) Suggest a formula for the complex ion present in **R**.

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For Examiner's Use

(iv) It is found that chromium still has an oxidation state of +3 in compound **S**. From this information and your knowledge of transition metal chemistry, suggest what has happened to cause the colour of the solution to change from green to violet.

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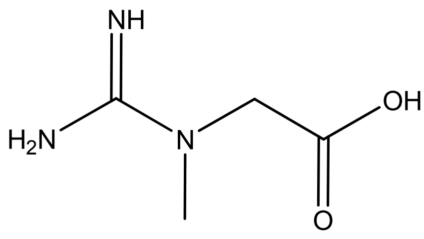
[6]

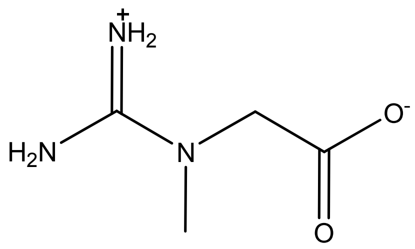
[Total: 15]

4 **(a)** Creatine is a nitrogenous organic acid that occurs naturally in vertebrates and helps to supply energy to all cells in the body, primarily muscle. Creatine supplements are used by athletes, bodybuilders, wrestlers, sprinters, and others who wish to gain muscle mass, typically consuming 2 to 3 times the amount that could be obtained from a very−high−protein diet.



creatine

Like amino acids, creatine exists in different ionised forms depending on the pH of the solution it is in. Creatine most commonly exists in the following form at pH 7.



creatine at pH 7

For Examiner's Use

(a) **(i)** In the boxes below, draw the most common form of creatine at pH 6 and pH 12.

pH 6

pH 12

(ii) Creatine can be used as a buffer at pH 7. Show, with the aid of an equation how it serves as a buffer at this pH when a small amount of base is added.

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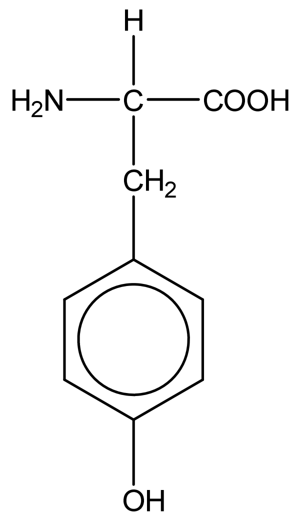
(iii) Amino acids like tyrosine can rotate plane polarised light.



tyrosine

Would you expect a molecule of creatine to rotate plane polarised light? Explain your answer.

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(iv) Suggest a simple chemical test to distinguish between creatine and tyrosine. State any observations you would make with each compound.

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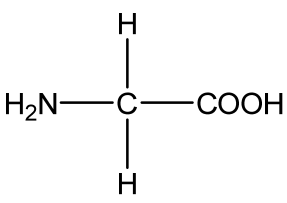
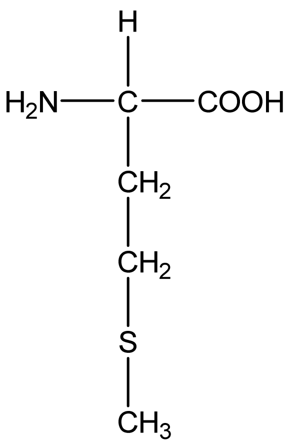
[7]

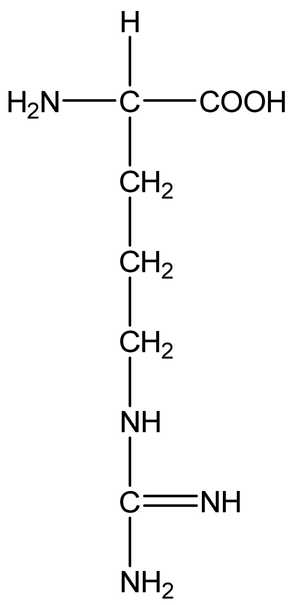
(b) Creatine is synthesised naturally in organisms from three amino acids: glycine, methionine and arginine.



glycine



methionine



arginine

(i) Draw the structural formula of the tripeptide Gly−Met−Arg.

For Examiner's Use

(ii) Name the type of reaction responsible for the formation of the tripeptide.

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A polypeptide chain consisting mostly of the three amino acids, glycine, methionine and arginine undergoes denaturation when heated.

(iii) What is meant by the term *denaturation*?

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(iv) Explain why the polypeptide undergoes denaturation upon heating.

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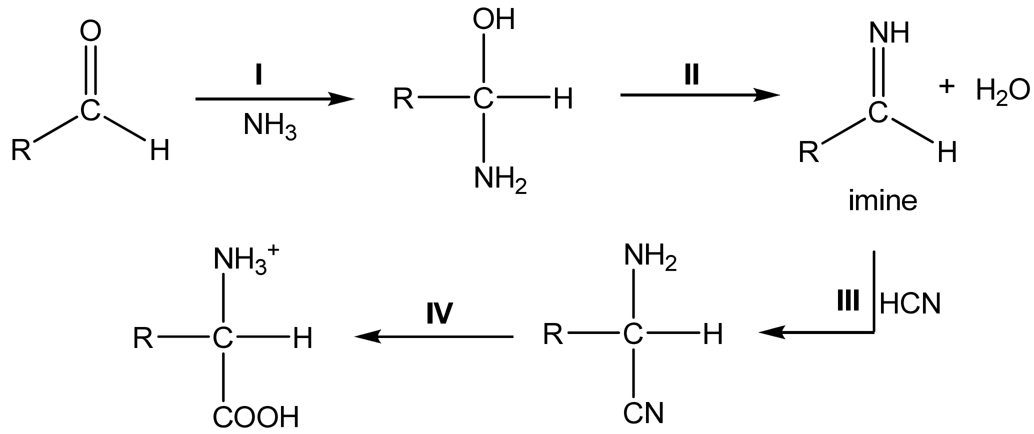
[4]

(c) Amino acids may be synthesised in the laboratory using the Strecker reaction. The reaction involves reacting an aldehyde with ammonia to form an imine which then reacts with HCN to give the amino acid.



(i) Suggest the structure of a suitable aldehyde for the synthesis of the amino acid methionine using the Strecker reaction.

For Examiner's Use

(ii) Identify the type of reaction for step **I** and step **II**.

step **I**: ....

step **II**: ....

(iii) Suggest suitable reagents and conditions for step **IV**.

reagents: ....

conditions: ....

(iv) Excess ammonia used in step **I** also reacts with $CH_3CH_2CH(CH_3)Br$ under ethanolic conditions in a sealed tube to give optically inactive products. Describe the mechanism of this reaction. In your answer you should show all charges and lone pairs of electrons and show the movement of electrons by curly arrows.

[7]

[Total: 18]

For Examiner's Use

5 Malonic acid, $HO_2CCH_2CO_2H$, is a diprotic acid with K_{a1} value of 1.41 x 10^{-3} mol dm^{-3} and K_{a2} value of 2.00 x 10^{-6} mol dm^{-3}.

(a) A student was instructed to titrate 25.0 cm^3 of 0.100 mol dm^{-3} $HO_2CCH_2CO_2H$, with 0.0625 mol dm^{-3} barium hydroxide, $Ba(OH)_2$.

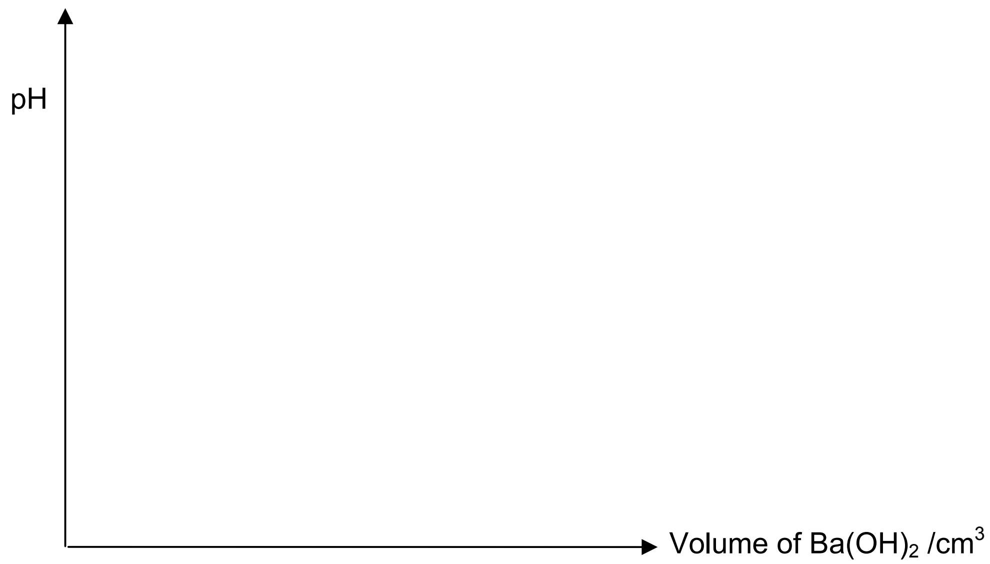
(i) Calculate the pH of 0.100 mol dm^{-3} malonic acid. (ignore the effect of pK_{a2} on the pH).

(ii) Calculate the volume of $Ba(OH)_2$ needed to completely react with 25.0 cm^3 of 0.100 mol dm^{-3} $HO_2CCH_2CO_2H$.

(iii) Calculate the pH of a solution which contains both $HO_2CCH_2CO_2H$ and $HO_2CCH_2CO_2^-$ in equal concentration.

(iv) Calculate the pH of a solution which contains both $HO_2CCH_2CO_2^-$ and $^-O_2CCH_2CO_2^-$ in equal concentration.

(v) Using your answers from **(a)(i)** to **(a)(iv)**, sketch the shape of the pH curve during the titration on the grid below.



[9]

For Examiner's Use

(b) A sample of hydrated sodium carbonate has the formula $Na_2CO_3.\boldsymbol{x}H_2O$. To determine the number of molecules of water of crystallisation, ***x***, 5.00 g of the hydrated salt was first dissolved in 250 cm^3 of deionised water. 25.0 cm^3 of the resultant solution was then pipetted out and titrated with 0.2 mol dm^{-3} of HC*l*.

(i) If 20.00 cm^3 of HC*l* was required for the titration, determine the concentration of the sodium carbonate in mol dm^{-3}.

(ii) Determine the value of ***x***.

[5]

[Total: 14]